



CSCE 633-700: Machine Learning

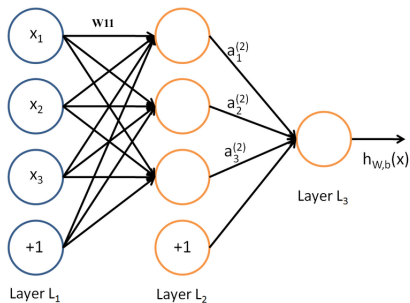
Neural Networks (Part 1)

Overview (5)

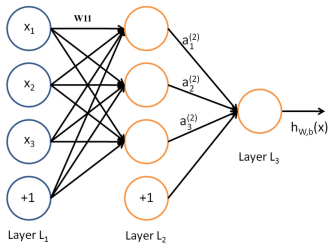
- Perceptron
 - Representation
 - Learning
 - Examples
- Multilayer Perceptron
 - Representation
 - Learning: Backpropagation
 - Practical issues

Multilayer Perceptron

- Type of feedforward neural network
- Can model non-linear associations
- “Multi-level combination” of many perceptrons



Multilayer Perceptron: Representation (1)



$$\alpha_1^{(2)} = f(W_{11}^{(1)} x_1 + W_{12}^{(1)} x_2 + W_{13}^{(1)} x_3 + b_1^{(1)})$$

$$\alpha_2^{(2)} = f(W_{21}^{(1)} x_1 + W_{22}^{(1)} x_2 + W_{23}^{(1)} x_3 + b_2^{(1)})$$

$$\alpha_3^{(2)} = f(W_{31}^{(1)} x_1 + W_{32}^{(1)} x_2 + W_{33}^{(1)} x_3 + b_3^{(1)})$$

$$h_{\mathbf{W},\mathbf{b}}(\mathbf{x}) = \alpha_1^{(3)} = f(W_{11}^{(2)} \alpha_1^{(2)} + W_{12}^{(2)} \alpha_2^{(2)} + W_{13}^{(2)} \alpha_3^{(2)} + b_1^{(2)})$$

Terminology

$W_{ij}^{(l)}$: connection between unit j in layer l to unit i in layer $l + 1$

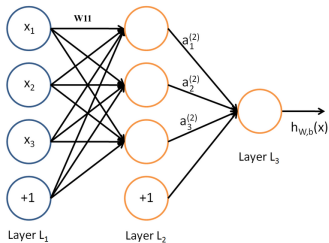
$\alpha_i^{(l)}$: activation of unit i in layer l

$b_i^{(l)}$: bias connected with unit i in layer $l + 1$

Forward propagation: The process of propagating the input to the output through the activation of inputs and hidden units to each node

Multilayer Perceptron: Representation (2)

Matrix notation



$$\alpha^{(2)} = f(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})$$

$$h_{W,b}(\mathbf{x}) = \alpha^{(3)} = f(\mathbf{W}^{(2)}\alpha^{(2)} + \mathbf{b}^{(2)})$$

$$\mathbf{W}^{(1)} = \begin{bmatrix} W_{11}^{(1)} & W_{12}^{(1)} & W_{13}^{(1)} \\ W_{21}^{(1)} & W_{22}^{(1)} & W_{23}^{(1)} \\ W_{31}^{(1)} & W_{32}^{(1)} & W_{33}^{(1)} \end{bmatrix}, \mathbf{b}^{(1)} = [b_1^{(1)} \ b_2^{(1)} \ b_3^{(1)}], \text{ etc.}$$

Multilayer Perceptron: Representation (3)

Alternative architectures

2 hidden layers, multiple output units

e.g. medical diagnosis: different outputs might indicate presence or absence of different diseases

